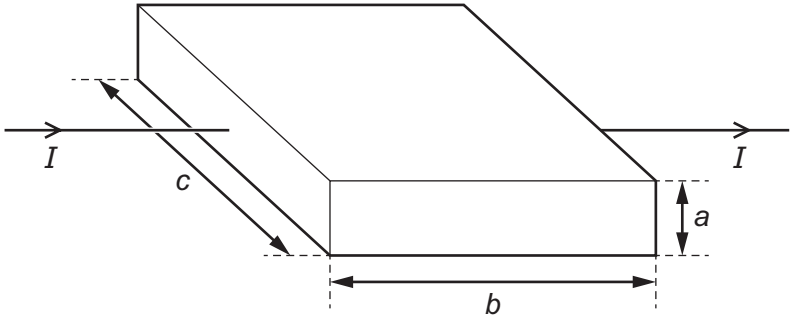
**31** The diagram shows a metal block.




The block has sides of length *a*, *b* and *c* as shown, and its volume is *V*. Each charge carrier has a charge –*q* and the number density of the charge carriers in the metal is *n*. It takes each charge carrier an average time of *t* to pass through the block.

What is an expression for the current *I*?

**A** $I = nqabc$ **B** $I = \frac{nqV}{t}$ **C** $I = \frac{nqbc}{t}$ **D** $I = nqaV$